Level 1 Functional Skills Mathematics
Sample
4748-119/219

www.cityandguilds.com
March 2021

Version 1.1

Mark scheme
March 2021

Level 1
SAMPLE

SAMPLE
4

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Guidance notes for Mark Schemes
Level 1 and Level 2

Notes for marking open response Problem Solving questions in Section 2:

The mark scheme has been carefully constructed to avoid penalising candidates repeatedly for similar errors.

1) The principle of follow through applies throughout unless otherwise stated. This allows the candidates to gain credit for subsequent correct calculation based on a previous incorrect answer. There is no follow-through between questions, but may be in multi-stage calculations within a question.

2) Units or numbers shown in brackets on the mark scheme are not required for the awarding of mark/s on the candidate's paper. However, if a candidate states units they must be correct:

eg 24(cm) means accept 24cm or 24 but not 24m

eg (£)72.5(0) means accept £72.50 or £72.5 or 72.50 or 72.5

3) Correct money format is expected in final answers unless otherwise indicated eg by brackets ie pounds must have two decimal places or no decimal places unless otherwise stated.

eg (£)5.00 or (£)5 not (£)5.0

eg (£)72.50 not (£)72.5

eg (£)37.43 not (£)37.432

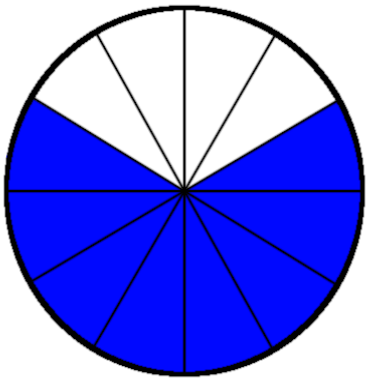
4) URT means unrounded, rounded or truncated; the underlining defines the acceptable limit of approximation:

eg 860. *8652* URT (U is the unrounded version)

the following are acceptable: 860 (T) or 861 (R) 860.8 (T) or 860.9 (R) or 860.86 (T) or 860.87 (R) or 860.865 (R) or 860.8652 (U) but not eg 900.

The 3rd and 4th columns of the mark schemes show the marks to be given for specific responses. Marks in bold are for fully correct answers. Where full marks are not achieved, examiners will award the marks that correspond to the responses given in the grey rows below. Any unforeseen but creditable responses are noted during the early stage of marking and are considered and, where appropriate, added to the mark scheme by the Chief Examiner when the mark scheme is finalised.

Where the marks are awarded for a *complete correct method with one calculation error*, examiners give the mark for a substantially correct solution with a single accuracy error or single (or consistent) early rounding, but not with a method error.

Maths Level 1 Sample 4: Section 1 – CALCULATOR NOT PERMITTED					
Candidates must not lose marks for incorrect spelling.					
Question	**Total marks**	**Marks**	**Marks awarded for**	**Item type**	**Subject content ref**
1	**1**	**1**	**35**	UPK Short answer fixed response	SCS7 [1]
2	**1**	**1**	**40(%)**	UPK Short answer fixed response	SCS16 [1]
3	**1**	**1**	**8.36**	UPK Short answer fixed response	SCS12 [1]
4	**1**	**1**	**1000**	UPK Short answer fixed response	SCS3 [1]
5	**1**	**1**	**125(cm³)**	UPK Short answer fixed response	SCS23 [1]
6	**1**	**1**	8 sectors of the circle shaded with at least one line of symmetry eg 	UPK Short answer fixed response	SCS24 [1]
7	**1**	**1**	**C**	UPK MC fixed response	SCS8 [1]
8	**1**	**1**	**28**	UPK Short answer fixed response	SCS4 [1]
9	**1**	**1**	**D**	UPK MC fixed response	SCS25 [1]
10	**1**	**1**	**205 030**	UPK Short answer fixed response	SCS1 [1]
11	**2**	**2**	**3.5**	Problem solving Short answer fixed response	SCS29 [2]
		1	28 for total seen **or** ÷ 8 seen		
12	**3**	**3**	**7.2(kg)**	Problem solving Short answer fixed response	SCS17 [3]
		2	7200(g)		
		1	720(g) grams for one loaf **or** correct use of proportion eg x 1.5 seen **or** complete correct method with one calculation or rounding error		
				Total for Section 1	**15 marks**

Maths Level 1 Sample 4: Section 2 – CALCULATOR PERMITTED					
Candidates must not lose marks for incorrect spelling. **NB** incorrect money format given as an answer should only be penalised **once** on the whole paper and will lose 1 mark. Do not penalise any subsequent incorrect format.					
Question	**Total marks**	**Marks**	**Marks awarded for**	**Item type**	**Subject content ref**
1	**1**	**1**	$\frac{7}{20}$	UPK Short answer fixed response	SCS16 [1]
2	**1**	**1**	**C($\frac{3}{4}$ full)**	UPK MC fixed response	SCS15 [1]
3	**1**	**1**	**D (-2ºC and 6ºC)**	UPK MC fixed response	SCS29 [1]
4	**1**	**1**	**20 009.6**	UPK Short answer fixed response	SCS11 [1]
5	**1**	**1**	**552**	UPK Short answer fixed response	SCS9 [1]
6	**1**	**1**	**no and valid explanation with reference to 140 miles** **eg 140 miles is too far to travel in 1 hour** **note: no calculation is expected**	Problem solving Short answer open response	Check [1] (SCS20)
7	**2**	**3**	**(£)86.25**	Problem solving Short answer open response	SCS14 [3]
		2	(£)2 070 for total to pay seen **or** complete correct method with one calculation or rounding error		
		1	method for finding 15% seen eg x 1.15 seen **or** ÷24 seen		
8	**3**	**3**	**1.775(m^2)**	Problem solving Short answer open response	SCS22 [3]
		2	1.3(m^2) **and** 0.475(m^2) **or** 0.8(m^2) **and** 0.975(m^2) **or** 2.535(m^2) **and** 0.76(m^2) **or** complete correct method with one calculation or rounding error		
		1	1.3(m^2) **or** 0.475(m^2) **or** 0.8(m^2) **or** 0.975(m^2) **or** 0.2535(m^2) **or** 0.76(m^2) **or** 1(m) **or** 0.8(m) for unlabelled dimensions		

<table>
<tr><td>9</td><td>3</td><td>3</td><td>$\frac{2}{5}$</td><td>Problem solving
Short answer
open response</td><td>SCS30 [1]
SCS31 [2]</td></tr>
<tr><td></td><td></td><td>2</td><td>$\frac{6}{15}$

or 6 out of 15 seen
or complete correct method with one calculation or rounding error</td><td></td><td></td></tr>
<tr><td></td><td></td><td>1</td><td>6 and 24 seen
or 15 seen for number of cards left
or 9 seen for the tally total</td><td></td><td></td></tr>
<tr><td>10</td><td>4</td><td>3</td><td>correct grouping of times with 3 equal group boundaries and correct totals or actual numbers for each group

eg

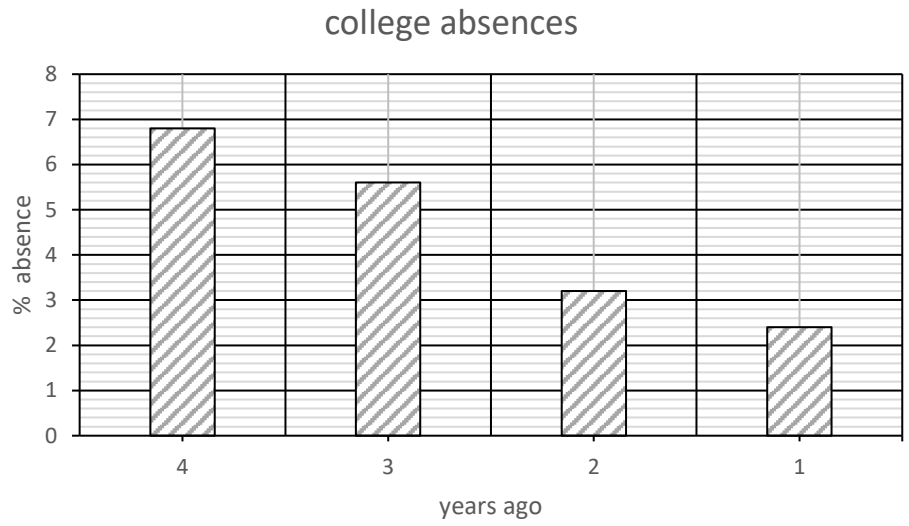
<table><tr><th>Time Length</th><th>0 – 5 minutes</th><th>6 – 10 minutes</th><th>11 - 15 minutes</th></tr><tr><td>Number in each group</td><td>8</td><td>4</td><td>6</td></tr></table>
eg

0 – 5 minutes 8
5 – 10 minutes 4
11 – 15 minutes 6</td><td>Problem solving
Short answer
open response</td><td>SCS28 [4]</td></tr>
<tr><td></td><td></td><td>2</td><td>all 18 times listed with correct grouping without boundaries stated
or correct grouping of all 18 times, but unequal group boundaries
or correct grouping of all 18 times with overlapping boundaries</td><td></td><td></td></tr>
<tr><td></td><td></td><td>1</td><td>some attempt to group times seen</td><td></td><td></td></tr>
<tr><td></td><td></td><td>1</td><td>valid comment based on their grouped data
eg there are more videos in the 0 – 5 minutes group
eg there are fewest videos in the 5 – 10 minutes group</td><td></td><td></td></tr>
</table>

11	**4**	**1**	**2(km) or 2000(m) seen for drone battery life**		SCS9 [1] SCS20 [1] SCS21 [2]
		2	**2.4(km) or 2400(m) seen for travel**		
		1	24 squares counted		
		1	**correct decision and valid explanation for their distance and their battery life** **eg no because 2.4km > 2km**		
12	**5**	**3**	**9699.354(litres)**	Problem solving Short answer open response	SCS5 [2] SCS6 [1] SCS17 [2]
		2	13.85622 seen for step 1 **or** 0.9699354 seen for step 2 **or** complete correct method with one calculation or rounding error		
		1	4.41 for r^2 **or** x 3.142 in step 1 **or** x 0.7 in step 2 **or** x 1000 in step 3		
		2	**decision consistent with their volume** **eg yes, 10 000 (litres in 5 hours) > 9 699 (litres)** **eg yes, (9699 ÷ 5 =) 1939(litres) < 2000(litres)**		
		1	their volume ÷ 5 (for number of litres per hour) or 10000(litres) seen for litres cleaned by machine in 5 hours		
13	**5**	**5**	**15:30 (time to leave home)**	Problem solving Short answer open response	SCS17 [1] SCS 20 [4]
		If 5 marks not awarded follow the 3 - part mark scheme			
		1	**17:15 (for arrival at check in)**		
		1	**17:00 (for arrival at car park)**		
		3	**correct time for leaving home consistent with time for arrival at car park / check in** **eg 15:30 (from 1hr 30 mins + 15:30 = 17:00)**		
		2	1 hour 30 minutes (for journey time from home to car park)		
		1	method for finding journey time eg 90 ÷ 60 = 1.5 hours (1 hour 30 minutes)		

14	**6**	**5**	***marks given here for acceptable methods based on yearly or monthly costs*** **(£)275.88 (total cost of joining plus the 50% health check) AND (£)300 (total cost of benefits without joining) for a year** **OR (£)22.99 for total cost of joining plus the 50% health check per month AND (£)25 for total cost of benefits without joining for a month from** **(£)60 ÷ 12 = (£)5 (£)42 ÷ 3 = (£)14 (£)36 ÷ 6 = (£)6**	Problem solving Short answer open response	SCS2 [1] SCS11 [1] SCS19 [1] SCS20 [3]
		4	(£)275.88 **or** (£)300 per year **or** (£)22.99 **or** (£)25 per month		
		3	(£)239.88 for annual subscription **and** (£)36 for reduced annual health check **or** (£)19.99 for monthly subscription **and** (£)3 for reduced monthly health check		
		2	(£)239.88 for annual subscription **or** (£)36 for reduced annual health check **or** (£)3 for reduced monthly health check **or** (£)60 **and** (£)168 **and** (£)72		
		1	÷ 12 (for monthly portion of annual injection) **or** x 12 (for annual subscription) **or** x 4 (for annual flea & worm treatment) **or** ÷ 3 (for monthly flea & worm treatment) **or** x 2 (for annual health check) **or** ÷ 6 (for monthly health check) **or** method for finding 50%		
		1	**decision consistent with their results AND explanation including figure(s)** **eg yes and (£)24.12 cheaper** **eg yes and (£)275.88 < (£)300**		
15	**6**	**2**	**2.4(%)**	Problem solving Short answer open response	SCS14 [2] SCS27 [4]
		1	0.024 seen **or** 948 ÷ 39 500		
		1	**bar chart with 4 bars for percentage absences AND suitable axis labels** **or line graph showing one line and 4 plots for percentage absences AND suitable axis labels in correct orientation** **(ie percentage on vertical scale)** ***note suitable title may be used to substitute or clarify axis label***		
		1	**bar chart: suitable continuous scale starting at (implied) zero and going to at least 6.8** **line graph: suitable continuous scales on both axes and going to at least 6.8** ***note line graph doesn't have to start at zero***		
		2	**bar chart: all 4 bar heights correct ± ½ small square** **line graph: all 4 points plotted correctly ± ½ small square**		
		1	bar chart: one bar height correct line graph: one point plotted correctly		
Total for Section 2 45 marks					

Example chart S2Q15



Example graph S2Q15

